

Zhuorui YUN

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Education

Guangdong Technion – Israel Institute of Technology, Guangdong, China

Aug 2023–Jul 2027

Bachelor of Science in Mechanical Engineering

- **Major:** Mechanical Engineering and Robotics
- **GPA:** 93.5/100 (Top 6%)
- **Relevant Coursework:** Algebra, Linear Systems, Dynamics, Introduction to Control, Introduction to Robotics, Mechatronics

Research & Project Experience

Quadcopter: Dynamics-Informed RL for a Monopedal Hopping Quadcopter

Jul 2025–Present

Undergraduate Researcher, Bio-inspired Mechatronics Laboratory

Training Pipeline & Policy Exploration

- Trained a direct estimated-state-to-motor PPO policy in Isaac Lab using 4,096 parallel environments, 37-dimensional observations, and 100 Hz four-motor control.
- Designed energy-manifold and efficiency reward shaping to coordinate active thrust with passive spring dynamics without a hopping state machine or low-level attitude PID.

Hardware-Aware Training Exploration

- Modeled measured platform dynamics, including 100–140 ms motor lag, 20–40 ms action delay, quadratic thrust mapping, and contact-rich stance/flight transitions.
- Built the custom 183-g STM32F103C8T6 flight platform with MPU6050 sensing and nRF24L01 communication for real-time hardware deployment.

Learning-Based Control & Results

- Deployed the ONNX policy at 100 Hz with 200 Hz eight-camera Vicon motion capture and a 15-state IMU–Vicon error-state Kalman filter for sim-to-real validation.
- Validated repeatable commanded-height hopping; representative hardware tests reduced cycle-averaged measured electrical power by 30.7% and normalized thrust by 49.8% versus tuned PID.

Publications

- R. Chen, Q. Zhang, Z. Zhong, **Z. Yun**, Y. Or, and M. Liu, “Dynamics-Informed Reinforcement Learning for Agile and Energy-Efficient Locomotion of a Monopedal Hopping Quadcopter,” *arXiv preprint*, arXiv:2609.15399, 2026.
- R. Chen, Q. Zhang, Z. Zhong, **Z. Yun**, Y. Or, and M. Liu, “Learning to Exploit Passive Dynamics for Energy-Efficient Target Hopping of a Spring-Legged Quadcopter,” *arXiv preprint*, arXiv:2609.15447, 2026. (**Under Review at IEEE RA-L**)

Robomaster KongFu Team, GTIIT

Jun 2025–Present

Algorithm Group Member & Electrical Control Team Lead, Sentry Robot

Embedded Communication & Real-Time Control

- Lead the electrical control subgroup for the sentry robot, coordinating embedded control development, actuator debugging, and system-level integration.
- Designed and debugged embedded control pipelines for real-time robot actuation, integrating onboard computation with STM32-based low-level control.
- Developed asynchronous UART/CAN communication pipelines between the onboard computer and microcontroller, supporting reliable command transmission and sensor feedback.
- Developed a ROS 2 Humble navigation stack integrating Livox MID360 3D LiDAR, point-cloud-to-LaserScan conversion, SLAM Toolbox, AMCL localization, and Nav2 planning.
- Built a configurable RMUL 2026 3v3 Gazebo simulation with URDF/Xacro robot description; implemented gimbal-sweep active sensing and scan-based fail-closed velocity guarding for navigation safety.

National Undergraduate Electronics Design Contest

Jul 31–Aug 4, 2026

Team Captain / Provincial Third Prize

- Led system integration and built K230/CanMV vision-based ball localization, transmitting coordinates via UART to the MSPM0G3507 controller for real-time servo actuation.

- Developed MSPM0G3507 firmware integrating 8-channel infrared sensing, encoders, motor drivers, OLED, and UART/I2C; implemented cascaded line-position/wheel-speed control with adaptive cornering, incremental PID, automatic lap-stop detection, and fault handling.

4-DOF Robotic Arm Teaching Pendant

Nov 2025–Dec 2025

Mechatronics Course Project / Arduino-based system integrating servo actuation, PWM control, and physical teaching input

System Integration & Hardware Debugging

- Built and debugged a 4-DOF robotic arm teaching system using Arduino Uno, MG996R high-torque servos, a PCA9685 16-channel PWM driver, potentiometers, and function buttons.
- Implemented real-time analog-read mapping from potentiometer voltage to servo angle commands, enabling the robotic arm to follow physical teaching input.
- Diagnosed unstable button readings caused by Arduino Uno pin 13 interference from the onboard LED circuit; reassigned the input to pin 8 and updated wiring/code to improve command stability.

Digital Filtering & Software Safety

- Implemented an exponential moving average (EMA) filter, $v_{\text{new}} = \alpha v_{\text{raw}} + (1 - \alpha)v_{\text{old}}$, to suppress analog input noise and reduce servo jitter.
- Added software safety constraints with Arduino `constrain()` limits to prevent servo commands from exceeding safe joint-angle ranges and damaging the mechanical structure.
- Built a one-click calibration workflow to rapidly align potentiometer polarity, servo direction, and joint zero positions during deployment and troubleshooting.

Reinforcement Learning Control of Underactuated Pendulum

Aug 2025

Discrete-action RL control for swing-up and stabilization under torque constraints

- Implemented and compared Q-learning, DQN, PPO, and TRPO for discrete-action pendulum swing-up and stabilization.
- Designed energy- and constraint-aware rewards; visualized learned policies in angle–angular-velocity space to assess convergence and control smoothness.

Technical & Language Skills

Programming & Numerical Computing

- **Python:** PyTorch, Stable Baselines3, NumPy, SciPy, reinforcement learning pipelines, physics simulation, data visualization
- **C/C++ & Embedded:** STM32 microcontrollers, UART/CAN communication, real-time control debugging, sensor integration
- **MATLAB & Simulink:** control modeling, dynamic systems analysis, pendulum simulation and comparison

Robotics & Dynamics Simulation Platforms

- **Isaac Sim / Isaac Lab:** DirectRLEnv-based RL training, parallel GPU simulation, domain randomization for sim-to-real transfer
- **MuJoCo:** 2D/3D contact-rich simulation, actuator modeling, closed-loop control validation
- **Hardware Tools:** Vicon motion capture, oscilloscopes, IMU/encoder sensors, soldering, robot hardware debugging

Mechanical Design & FEA

- **CAD & 3D Modeling:** SolidWorks, AutoCAD, Fusion 360; mechanical part modeling, assembly design, and engineering drawing preparation
- **Finite Element Analysis:** ANSYS-based structural simulation and stress/deformation analysis for mechanical components

Languages

- Chinese (native)
- English (professional working proficiency)
- **TOEFL iBT:** 100/120 (CEFR C1), Aug 2026

Positions of Responsibility

Undergraduate Peer Tutor, Department of Mechanical Engineering, GTIIT Jun 2024–Oct 2024

- Led a study group of 10 incoming students during the preparatory semester, providing weekly sessions on study planning and subject-specific guidance.
- Facilitated themed experience-sharing meetings each week, covering exam strategies, course selection, and academic adjustment.

Student Representative, Mechanical Engineering Program, GTIIT Oct 2024–Present

- Represent the student body in semesterly meetings with Prof. Varennyam Achal (Deputy Dean of Undergraduate Studies); gather and relay student feedback on curriculum design, examination arrangements, and campus life to inform faculty decisions.

Honors, Awards & Scholarships

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| • Vice Chancellor’s List, GTIIT — Top 5% in program | 2024–2025 |
| • Dean’s List, GTIIT — Top 10% of the cohort | 2023–2024 |
| • Most Innovative Project Award, Research Poster Competition, GTIIT | Apr 2026 |
| • Provincial Third Prize, National Undergraduate Electronics Design Contest (NUEDC) | Aug 2026 |
| • Chancellor’s Scholarship (Third Class), awarded for academic excellence | 2024, 2025 |

URLs

GitHub: <https://github.com/ZhuoruiYUN>
LinkedIn: <https://www.linkedin.com/in/zhuorui-yun-5675aa2b9/>
Portfolio: <https://zhuoruiyun.github.io/>
arXiv: <https://arxiv.org/abs/2609.15399> <https://arxiv.org/abs/2609.15447>